ON-BOARD DIAGNOSIS SYSTEM [ADVANCED KEYLESS ENTRY SYSTEM]

id091400111233

Outline

- The on-board diagnostic function consists of the following functions:
  - — A malfunction detection function, which detects overall malfunctions in the advanced keyless entry system-related parts.
  - — A memory function, which stores detected DTCs.
  - — A display function, which indicates malfunction locations and status via DTC output.
  - — A PID/data monitoring function, which reads out specific input/output signals and verifies the input/output condition.
- Using the Mazda Modular Diagnostic System (M-MDS), DTCs can be read out and deleted, and the PID/data monitoring function can be activated.

Malfunction detection function

- Detects malfunctions in input/output signals.
- If a malfunction occurs, the start stop unit records the malfunction as a DTC. A recorded DTC can be read by the Mazda Modular Diagnostic System (M-MDS).

DTC table

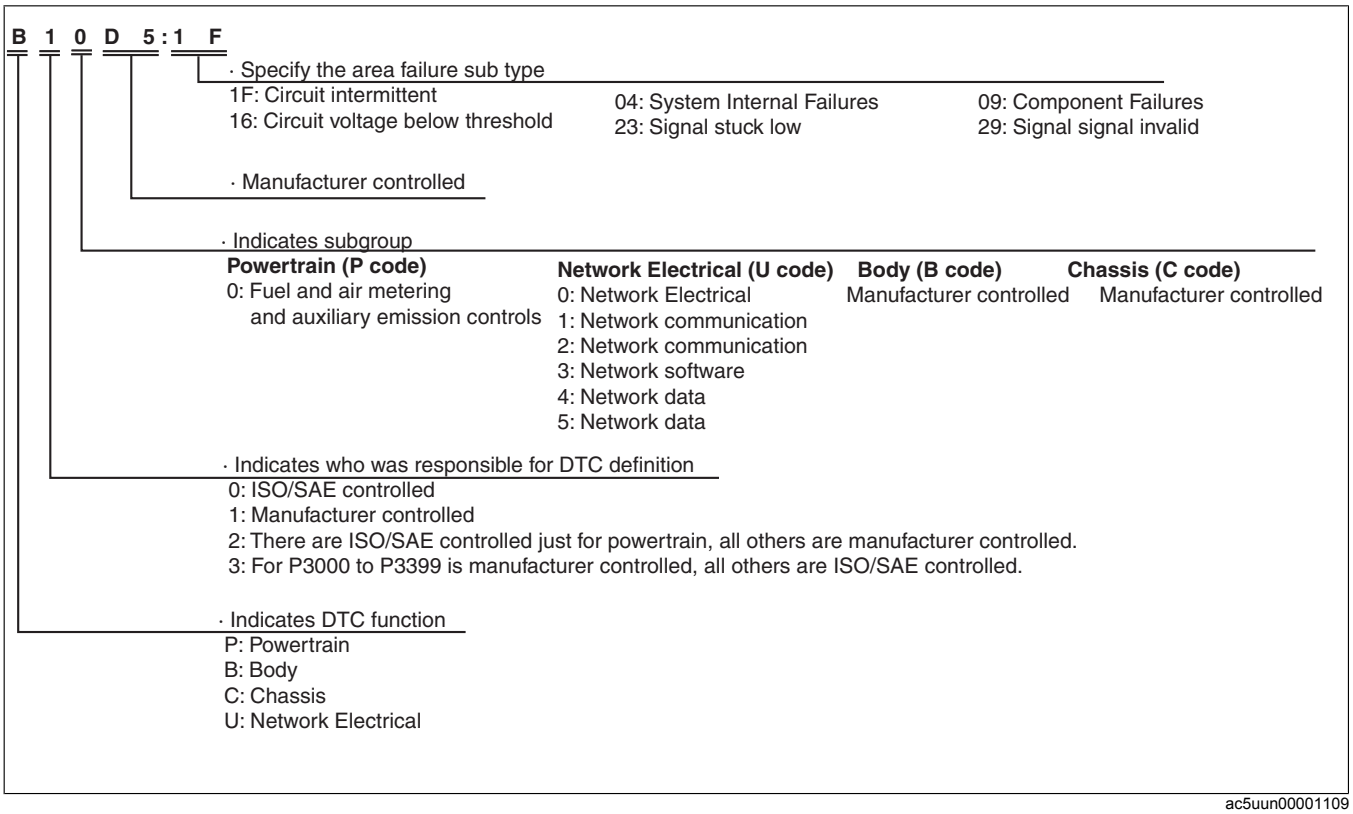
×: Applicable
—: Not applicable

DTC No.	KEY warning light (red)	Description	Fail-safe	Drive cycle	Self test type*1	Memory function
B10C6:1F	On	Keyless antenna (exterior, rear) circuit malfunction	×	—	C, D	×
B10C7:1F	On	Keyless antenna (interior, rear) circuit malfunction	×	—	C, D	×
B10C9:1F	On	Keyless antenna (interior, front) circuit malfunction	×	—	C, D	×
B10D1:23	On	Request switch (LF) circuit malfunction	—	—	C	×
B10D3:23	On	Request switch (RF) circuit malfunction	—	—	C	×
B11FD:1F	On	Keyless antenna (exterior, LF) circuit malfunction	×	—	C, D	×
B1210:1F	On	Keyless antenna (exterior, RF) circuit malfunction	×	—	C, D	×
B13C3:04	On	LF control unit internal malfunction	×	—	C	×
B13C3:09	On	LF control unit malfunction	×	—	C	×
B13C3:16	On	LF control unit power supply voltage low input	×	—	C	×
B13C3:29	On	Communication error between start stop unit and LF control unit	×	—	C	×

*1 : C: CMDTC self test, D: ODDTC self test

DTC 7-digit code definition

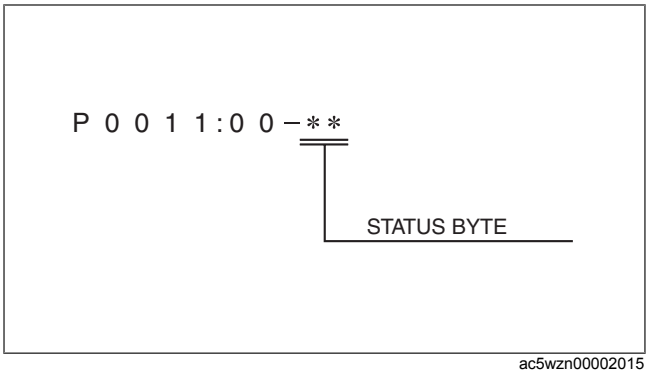
- When related systems or components have failed, the CM stores the DTC of the malfunctioning part in the CM memory, and allows for the retrieval of the store data using scanning tool when necessary. The DTCs are indicated using seven digits. Each digit indicates the following.



ac5uun00001109

Status byte for DTC

- The two digits (two digits after hyphen (-)) after the seven-digit DTC.
- A code which indicates the pending code, current/ past malfunction status, or warning illumination status.
- Can be read by performing a CMDTC self-test using the Mazda Modular Diagnostic System (M-MDS).
- For details on the status byte, refer to the explanation on the Mazda Modular Diagnostic System (M-MDS) when reading the DTC.



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Detection condition for the applicable DTC

DTC No.	Description	Detection condition
B10C6:1F	Keyless antenna (exterior, rear) circuit malfunction	• Start stop unit detects an open circuit or short to ground in the keyless antenna (exterior, rear) circuit with the ignition switched off.
B10C7:1F	Keyless antenna (interior, rear) circuit malfunction	• Start stop unit detects an open circuit or short to ground in the keyless antenna (interior, rear) circuit with the ignition switched off.
B10C9:1F	Keyless antenna (interior, front) circuit malfunction	• Start stop unit detects an open circuit or short to ground in the keyless antenna (interior, front) circuit with the ignition switched off.
B10D1:23	Request switch (LF) circuit malfunction	• Start stop unit detected any of the following conditions at a vehicle speed of **5 km/h {3 mph} or more**. — The request switch (LF) is continuously ON for **2 min or more**. — The request switch (LF) was ON **7 times** before the vehicle speed decreased to **less than 5 km/h {3 mph}**.

DTC No.	Description	Detection condition
B10D3:23	Request switch (RF) circuit malfunction	• Start stop unit detected any of the following conditions at a vehicle speed of **5 km/h {3 mph} or more**. — The request switch (RF) is continuously ON for **2 min or more**. — The request switch (RF) was ON **7 times** before the vehicle speed decreased to **less than 5 km/h {3 mph}**.
B11FD:1F	Keyless antenna (exterior, LF) circuit malfunction	• Start stop unit detects an open circuit or short to ground in the keyless antenna (exterior, LF) circuit with the ignition switched off.
B1210:1F	Keyless antenna (exterior, RF) circuit malfunction	• Start stop unit detects an open circuit or short to ground in the keyless antenna (exterior, RF) circuit with the ignition switched off.
B13C3:04	LF control unit internal malfunction	• Start stop unit detects the LF control unit internal malfunction.
B13C3:09	LF control unit malfunction	• When any of the following conditions is met: — Start stop unit detects that the LF state signal of the LF control unit is low for **5 s or more**. — With the communication between the start stop unit and LF control unit being performed normally, there is no response from the LF control unit even though there is the signal transmission request from the start stop unit to the LF control unit, and the LF state signal becomes low.
B13C3:16	LF control unit power supply voltage low input	• Start stop unit power supply circuit voltage of **8.5 V or more or less than 16.5 V** and LF control unit power supply circuit voltage of **5 V or more or less than 8.5 V** are detected for **5 s or more**.
B13C3:29	Communication error between start stop unit and LF control unit	• Start stop unit detects the communication error with the LF control unit **10 times**.

Data monitor function

- With the PID/data monitor function, the input/output signal monitor items set in the start stop unit can be selected and read out in real-time.
- The items which can be selected are as follows.

PID	Unit/ Operation	Data contents	Data read/use method	Module control terminal
LG/T_LK_SW *1	Off/On	Request switch (liftgate) operation condition • Off: Request switch (liftgate) is off. • On: Request switch (liftgate) is on.	The request switch (liftgate) input is an input signal to the start stop unit through the LF control unit.	2F (LF control unit communication)
RQ_SW_LF	Off/On	Request switch (LF) operation condition • Off: Request switch (LF) is off. • On: Request switch (LF) is on.	The signal is input to the start stop unit through the LF control unit of the request switch (LF).	2F (LF control unit communication)
RQ_SW_RF	Off/On	Request switch (RF) operation condition • Off: Request switch (RF) is off. • On: Request switch (RF) is on.	The signal is input to the start stop unit through the LF control unit of the request switch (RF).	2F (LF control unit communication)
T_SEN_LF	Off/On	Displays in the Mazda Modular Diagnostic System (M-MDS) but it does not operate.		
T_SEN_RF	Off/On	Displays in the Mazda Modular Diagnostic System (M-MDS) but it does not operate.		

*1 : 5HB

Simulation function

- The simulation function drives the output parts set in the start stop unit regardless of the start stop unit control condition.
- The items which can be selected are as follows.

Note

- Simulation items on parts which are not equipped to the vehicle do not operate.

Simulation item	Unit/ Operation	Data contents	Data read/use method	Output part name	Operation condition
BZR_OUT	Off/On	• Off: Keyless beeper does not sound. • On: Keyless beeper sounds.	• When the simulation item BZR_OUT is turned on, a force override signal is sent to the LF control unit from the start stop unit, and the LF control unit drives (sounds) the keyless beeper. • With adoption of a separate excitation buzzer to the keyless beeper, buzzer does not sound normally even if the voltage is applied to the buzzer unit because there is no oscillation circuit. The integrity of the keyless beeper is determined using the simulation item BZR_OUT. • If the keyless buzzer does not sound even though the simulation function BZR_OUT is on, perform diagnosis using the following procedure. — Press the remote transmitter lock/unlock button repeatedly and measure the terminal voltage of the keyless buzzer when the button is pressed. **Voltage is output** • Keyless buzzer malfunction may have occurred. **Voltage is not output** • Related wiring harness, or LF control unit malfunction may have occurred.	Keyless beeper	Ignition switched ON (engine off or on)
T_SEN_P_LF	Off/On	**Note** • Displayed on Mazda Modular Diagnostic System (M-MDS) but does not function.	—	—	—
T_SEN_P_RF	Off/On	**Note** • Displayed on Mazda Modular Diagnostic System (M-MDS) but does not function.	—	—	—